

7. (a) A student is asked to prepare a standard solution of sodium carbonate of concentration $0.0500 \text{ mol dm}^{-3}$.

The first step involved in the preparation of a standard solution is to weigh the appropriate mass of solid in a weighing bottle.

Calculate the mass of sodium carbonate required and describe the remaining steps that the student should take to prepare 250 cm^3 of this standard solution. [6]

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- (b) The standard solution prepared in part (a) was used to determine the concentration of dilute hydrochloric acid.

25.0 cm³ samples of the sodium carbonate solution were titrated against the hydrochloric acid. These were the results.

Titration	1	2	3	4
Final reading / cm ³	26.50	26.80	26.20	26.55
Initial reading / cm ³	0.40	0.15	0.00	0.25
Titre / cm ³	26.10	26.65	26.20	26.30

- (i) Calculate the mean titre that should be used to determine the concentration of the hydrochloric acid. [1]

Mean titre = cm³

- (ii) The burette used in the titrations has an uncertainty for each reading of $\pm 0.05 \text{ cm}^3$. Estimate the maximum percentage error in the **titre** in titration 4. Show your working. [1]

Percentage error = %

- (iii) Apart from errors in reading the burette, suggest **one** reason why incorrect titres may have been obtained when carrying out the titrations. Explain the effect of this error on the value of the titre obtained. [2]

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- (c) A student wanted to verify that the ammonium sulfate content in a fertiliser was as given on the packet.

He boiled 4.24 g of the fertiliser with 50.0 cm³ of 1.00 mol dm⁻³ sodium hydroxide solution until no more ammonia was evolved.

The residual solution containing excess sodium hydroxide was made up to 250 cm³ with distilled water in a standard volumetric flask. A 25.0 cm³ portion of this solution required 22.65 cm³ of 0.090 mol dm⁻³ hydrochloric acid for neutralisation.

- (i) Calculate the number of moles of sodium hydroxide in the 250 cm³ flask and hence the number of moles that reacted with the fertiliser. [3]

$n(\text{NaOH}) = \dots\dots\dots \text{mol}$

- (ii) The equation for the reaction between ammonium sulfate and sodium hydroxide is:



Use your answer to part (i) to calculate the percentage by mass of ammonium sulfate in the fertiliser. [2]

Percentage = $\dots\dots\dots\%$

